Sedentary time predicts an increased annual risk of transitioning from healthy to burnout-related coping patterns

No significant effect of phyiscal activity on coping transitions

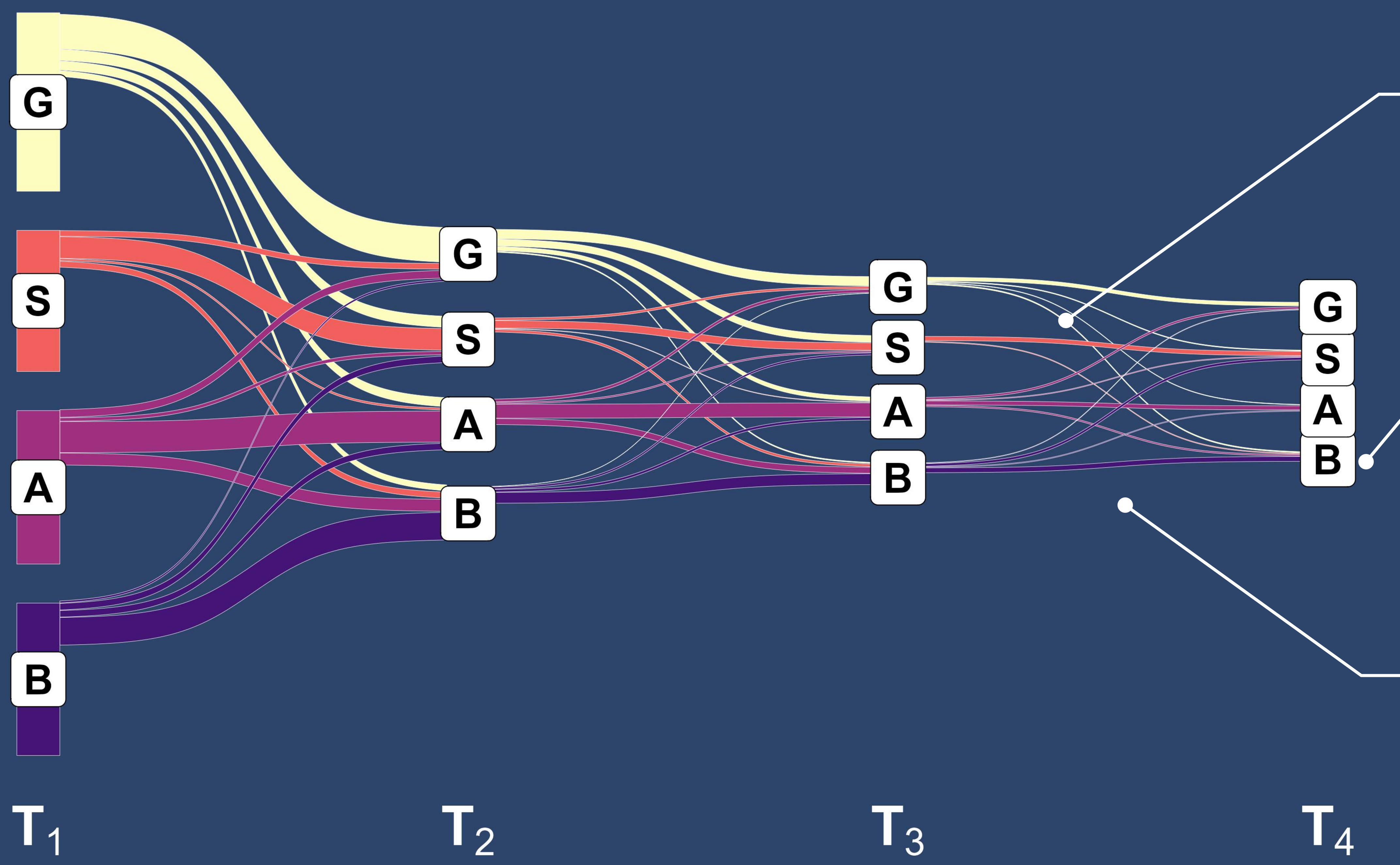


Figure 1. Proportion of students in each coping pattern at T_1–T_4 (person-centered time points aligned with each participant's first participation between 2022 - 2025), indicated by node widths, and their proportional transitions across consecutive years, indicated by flow widths.

Results

1 Sedentary time
Higher sedentary time predicts transitions from healthy (G) to burnout-related coping (B) within one academic year (HR = 2.30, 95% CI: 1.15 – 4.59).

2 Transition probability
The probability of B (Burnout) being the next patterns after G (Good health) was 3% (95% CI: 1% – 22%) for low (-1 SD) and 16% (95% CI: 6% – 38%) for high (+1 SD) sedentary time.

3 Physical activity
Physical activity was not significantly associated with transition hazard ratios between coping patterns (HRs = 0.77 – 1.29, all 95% CIs included 1).

Research question

How are physical activity (PA) and sedentary time (ST) associated with transitions between academic coping behavior and experience patterns in students over time?

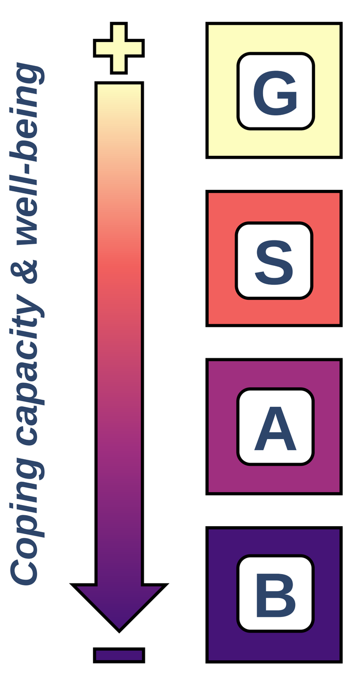
Design

Longitudinal cohort survey study (2022 – 2025)

N_{T1} = 785 → N_{T2} = 677 → N_{T3} = 341 → N_{T4} = 162

Method

Outcome: **Academic coping behavior and experience patterns**



Good health: Health promoting attitude to work, optimal professional motivation, no health risks

Sparing personal investment: Resource efficient attitude to work, protection, no health risks

Ambitious: Excessive engagement, workaholism, over-motivation, vulnerability to health risks

Burnout: Burnout syndrome, low motivation, resignation, high vulnerability and health risks

(WCEP, Schaarschmidt & Fischer, 1997, 2008)

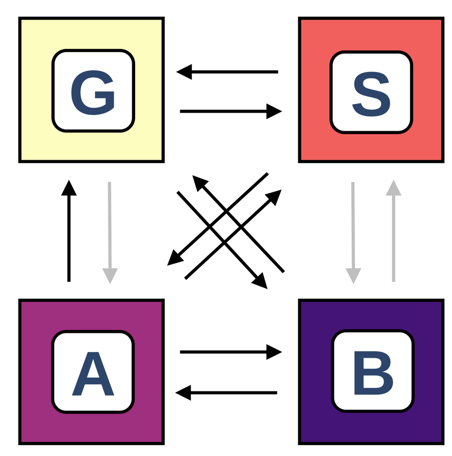
Predictors

PA: Moderate to vigorous intensity PA in metabolic equivalent of task minutes per week (IPAQ, Craig et al., 2003)

ST: Daily sitting time in minutes (IPAQ, Craig et al., 2003)

Data analysis

Continuous-time multi-state Markov models were applied to estimate the effects of PA and ST on transition probabilities between patterns over time.



When Sitting Still Leads to a Standstill: The Role of Physical Activity and Sedentary Time in Coping with Academic Stress

Ludwig Piesch[1], Katrin Obst[1], Susen Köslich-Strumann[1], & Till Utesch[1]
[1] *University of Luebeck*

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